Orthogonal Vectors and Subspaces

Orthogonal means perpendicular, and in markets it means "has nothing to do with". A market-neutral trade is orthogonal to the market; uncorrelated returns are orthogonal vectors. And the four subspaces of every matrix come in two perpendicular pairs.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. **Conditions and important concepts**
   12 short definitions: dot product, length, orthogonal subspaces, complements, A^TA, Fredholm's alternative.
2. **Procedure**
   Test orthogonality; find an orthogonal complement.
3. **2 × 2 example**
   Nifty and its twin: two perpendicular pairs.
4. **3 × 3 example**
   Nifty, bonds, balanced: the do-nothing trade is perpendicular to every rule.
5. **Bigger example: returns as vectors**
   Correlation is a cosine; a market-neutral strategy is orthogonal to the market. With a correlation explorer.
6. **Connection: the big picture**
   The fundamental theorem, part 2, and the road to projections.
7. **Final memory table**
   Everything on one screen.

METHODS IN THIS LESSON: Orthogonality test $x^Ty = 0$ · Length $\|x\|$ · Orthogonal subspaces · Orthogonal complements · Correlation as cosine · $N(A^TA) = N(A)$ · Fredholm's alternative

PART 2 Prerequisites

Click any link to review it, then come back.

- The four subspaces and the hidden link (Lesson 10).

- The dot product x^Ty and R^TR and covariance (Lesson 5).
- Loops and arbitrage (Lesson 12).

PART 3 **Explanation**

THE NUMBERS USED IN THE EXAMPLES

Fund matrix (rows: budget, return, exposure): Nifty $(1, 12, 1)$, Nifty ETF $(1, 12, 1)$, bonds $(1, 7, 0)$, balanced $(1, 9.5, 0.5)$.

Scenario returns (bad, normal, good year), as in Lesson 5: Nifty $(-10, 10, 30)$, average 10; midcap $(-16, 12, 34)$, average 10; a market-neutral strategy $(8, -7, 8)$, average 3. **Demeaned** (each minus its own average): Nifty $(-20, 0, 20)$, midcap $(-26, 2, 24)$, strategy $(5, -10, 5)$.

1. Conditions and important concepts

1. Orthogonal vectors

RULE · THE ORTHOGONALITY TEST

Two vectors are **orthogonal** (perpendicular, at 90°) exactly when their dot product is zero:

$$x \perp y \iff x^Ty = x_1y_1 + x_2y_2 + \cdots + x_ny_n = 0$$

Example: the zero-cost trade $(1, -1)$ and the budget row $(1, 1)$: $1 \cdot 1 + (-1) \cdot 1 = 0$. Perpendicular.

2. Length

DEFINITION · LENGTH (NORM)

$$\|x\|^2 = x^Tx = x_1^2 + \cdots + x_n^2 \qquad \|x\| = \sqrt{x^Tx}$$

Nifty's demeaned returns $(-20, 0, 20)$: $\|x\|^2 = 400 + 0 + 400 = 800$. For a demeaned return vector, $\|x\|^2/3$ is the variance, so length measures **risk**.

3. Why the test works: Pythagoras

x and y are perpendicular exactly when they form a right triangle with $x + y$:

$$\|x\|^2 + \|y\|^2 = \|x + y\|^2 = (x + y)^T(x + y) = \|x\|^2 + \|y\|^2 + 2x^T y$$

The two sides agree exactly when $2x^T y = 0$, i.e. $x^T y = 0$.

4. The zero vector

$0^T y = 0$ for every y, so the zero vector is orthogonal to everything.

5. Orthogonal subspaces

> **DEFINITION · ORTHOGONAL SUBSPACES**
> Subspace S is orthogonal to subspace T when **every** vector in S is orthogonal to **every** vector in T.

A wall and the floor are **not** orthogonal subspaces: the line where they meet lies in both, and a non-zero vector is never perpendicular to itself. Two lines through the origin at right angles are orthogonal subspaces.

6. Row space ⊥ null space

$Ax = 0$ says: row $1 \cdot x = 0$, row $2 \cdot x = 0$, ..., every row of A is perpendicular to x. Then every **combination** of rows is too. So:

$\boxed{C(A^T) \perp N(A)}$ In money terms: a do-nothing trade is perpendicular to every rule.

7. Column space ⊥ left null space

The same argument for A^T: $A^T y = 0$ makes y perpendicular to every column of A, and so to every combination of columns.

$\boxed{C(A) \perp N(A^T)}$ In money terms: every reachable target is perpendicular to the hidd

8. Orthogonal complements

RULE · ORTHOGONAL COMPLEMENTS (FUNDAMENTAL THEOREM, PART 2)

Not only are they perpendicular, their dimensions add up to the whole space: $r + (n - r) = n$ and $r + (m - r) = m$. So $N(A)$ contains **every** vector perpendicular to the row space, and $N(A^T)$ contains **every** vector perpendicular to the column space. Each is the **orthogonal complement** of the other.

9. Hedged = orthogonal

A position's market exposure is (exposure row) · (position). So a position is **market-neutral** exactly when it is orthogonal to the exposure row. Lesson 5's zero-cost, market-neutral trade $(-6, 1, 5)$ is orthogonal to both the budget row $(1, 1, 1)$ and the exposure row $(1, 0, 1.2)$: $-6 + 1 + 5 = 0$ and $-6 + 0 + 6 = 0$.

10. Correlation is a cosine

RULE · CORRELATION = COSINE OF THE ANGLE

For demeaned return vectors x and y: $\cos\theta = \dfrac{x^T y}{\|x\|\,\|y\|}$ is exactly their **correlation**.

Orthogonal ($x^T y = 0$) means **uncorrelated**.

11. AᵀA: when is it invertible?

RULE · N(AᵀA) = N(A)

$A^T A$ is square ($n \times n$) and symmetric. It has the same null space and the same rank as A, so **$A^T A$ is invertible exactly when the columns of A are independent.** (Proof: if $A^T Ax = 0$ then $x^T A^T Ax = \|Ax\|^2 = 0$, so $Ax = 0$.) Lesson 14 needs this.

12. Fredholm's alternative: a certificate of "no solution"

RULE · FREDHOLM'S ALTERNATIVE

Exactly one of these is true: (a) $Ax = b$ has a solution, or (b) some y has $A^T y = 0$ and $y^T b = 1$. Multiplying the equations by y and adding gives $0 = 1$: proof that no solution exists.

In markets: Lesson 12's loop $y = (1, 1, -1)$ with spread quotes $(100, 150, 240)$ gives $A^T y = 0$ and $y^T e = 10$. Scale it to $\frac{1}{10}y$ and $y^T e = 1$: a certificate that the quotes can't come from real prices, i.e. an **arbitrage**.

2. Procedure

A. Are two vectors (or subspaces) orthogonal?

Step	What to do
1	Two vectors: compute $x^T y$. Zero ⇒ orthogonal.
2	Two subspaces: take a basis of each and check every pair of basis vectors. All zero ⇒ orthogonal.

B. The orthogonal complement of a subspace

Step	What to do
1	Put a basis of the subspace as the **rows** of a matrix B.
2	The complement is $N(B)$: solve $Bx = 0$ with the recipe from Lesson 7.
3	Check: dimensions add up to n.

3. 2 × 2 example

Nifty and its twin with budget and return: $A = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}$, rank 1 (Lesson 10).

Pair	Bases	Dot product	Dimensions
Row space ⊥ null space (in $\mathbb{R}^2$)	$(1, 1)$ and $(-1, 1)$	$1(-1) + 1(1) = 0$ ✓	$1 + 1 = 2 = n$
Column space ⊥ left null space (in $\mathbb{R}^2$)	$(1, 12)$ and $(-12, 1)$	$1(-12) + 12(1) = 0$ ✓	$1 + 1 = 2 = m$

Meaning. The row space measures "total Nifty held"; the do-nothing trade (swap one twin for the other) is perpendicular to it, so it doesn't change that measurement. The reachable targets $(B, 12B)$ are perpendicular to the hidden link $(-12, 1)$: return $= 12\times$ budget.

4. 3 × 3 example

Nifty, bonds, balanced: $A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix}$, rank 2. From Lesson 10: row space basis $(1, 0, 0.5)$, $(0, 1, 0.5)$; null space $(-0.5, -0.5, 1)$; column space $(1, 12, 1)$, $(1, 7, 0)$; left null space $(7, -1, 5)$.

Row space ⊥ null space

- $(1, 0, 0.5) \cdot (-0.5, -0.5, 1) = -0.5 + 0 + 0.5 = 0$ ✓
- $(0, 1, 0.5) \cdot (-0.5, -0.5, 1) = 0 - 0.5 + 0.5 = 0$ ✓
- Also each original rule: budget $(1, 1, 1) \cdot (-0.5, -0.5, 1) = 0$, return $(12, 7, 9.5) \cdot (-0.5, -0.5, 1) = -6 - 3.5 + 9.5 = 0$, exposure $(1, 0, 0.5) \cdot (-0.5, -0.5, 1) = 0$ ✓.
- Dimensions: $2 + 1 = 3 = n$. The null space is the **whole** line perpendicular to the plane of rules.

Column space ⊥ left null space

- Nifty $(1, 12, 1) \cdot (7, -1, 5) = 7 - 12 + 5 = 0$ ✓
- Bonds $(1, 7, 0) \cdot (7, -1, 5) = 7 - 7 + 0 = 0$ ✓
- Dimensions: $2 + 1 = 3 = m$. The plane of reachable targets $R = 7B + 5E$ is exactly "everything perpendicular to $(7, -1, 5)$".

A target b is reachable $\iff b \perp (7, -1, 5)$. $(10, 105, 7)$: $70 - 105 + 35 = 0$ ✓. $(10, 105, 5)$: $70 - 105 + 25 = -10$ ✗. Lesson 14 asks: when b isn't perpendicular, what's the **closest** reachable target?

> ▼ **Check yourself (MIT lecture): for $A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \end{bmatrix}$, describe the row space and the null space.**
>
> Row 2 = 2 × row 1, so the row space is the line through $(1, 2, 5)$ (dimension 1). The null space is the plane $x + 2y + 5z = 0$ (dimension 2), with basis $(-2, 1, 0)$ and $(-5, 0, 1)$. Check: $(1, 2, 5) \cdot (-2, 1, 0) = 0$ and $(1, 2, 5) \cdot (-5, 0, 1) = 0$ ✓. A line and a plane, perpendicular, $1 + 2 = 3$.

5. Bigger example: returns as vectors

Treat each asset's demeaned scenario returns as a vector in $\mathbb{R}^3$:

$$\text{Nifty } a = \begin{bmatrix} -20 \\ 0 \\ 20 \end{bmatrix} \qquad \text{midcap } m = \begin{bmatrix} -26 \\ 2 \\ 24 \end{bmatrix} \qquad \text{strategy } s = \begin{bmatrix} 5 \\ -10 \\ 5 \end{bmatrix}$$

Step 1: the strategy is orthogonal to the market

$$a^T s = (-20)(5) + (0)(-10) + (20)(5) = -100 + 0 + 100 = \boxed{0}$$

Orthogonal, so **uncorrelated** with Nifty: a market-neutral strategy. It makes money in normal years and loses in extremes, with no tilt to the market's direction.

Step 2: midcap is almost parallel to Nifty

$$a^T m = 520 + 0 + 480 = 1000 \qquad \|a\|^2 = 800 \qquad \|m\|^2 = 676 + 4 + 576 = 1256$$

$$\cos\theta = \frac{1000}{\sqrt{800}\,\sqrt{1256}} = \frac{1000}{28.28 \times 35.44} \approx 0.998$$

Correlation 0.998: the angle between them is only about 4°.

Step 3: AᵀA, the "overlap table"

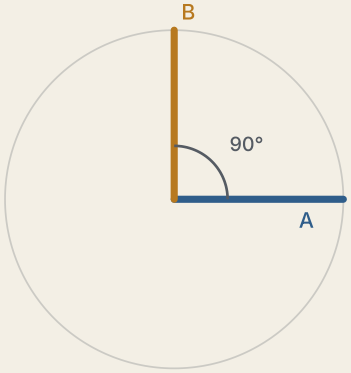
With Nifty and bonds as the columns of the 3-rule fund matrix, $A^T A = \begin{bmatrix} 1+144+1 & 1+84+0 \\ 85 & 1+49+0 \end{bmatrix} = \begin{bmatrix} 146 & 85 \\ 85 & 50 \end{bmatrix}$, with $146 \times 50 - 85^2 = 7300 - 7225 = 75 \neq 0$: invertible (independent columns). With Nifty and its twin, $A^T A = \begin{bmatrix} 146 & 146 \\ 146 & 146 \end{bmatrix}$, with $146^2 - 146^2 = 0$: singular, because the columns are dependent.

Try it: correlation as an angle

Asset A returns (%): bad -10 normal 10 good 30

Asset B returns (%): bad 8 normal -7 good 8

Preset choose...



Demeaned: $x = (-20, 0, 20), y = (5, -10, 5)$.

$x^T y = (-20)(5) + (0)(-10) + (20)(5) = 0$

$\|x\| = 28.2843$, $\|y\| = 12.2474$, so correlation $= \cos\theta = 0$, angle $\approx 90°$.

Orthogonal: uncorrelated.

6. Connection: the big picture

$$\underbrace{\mathbb{R}^n}_{\text{trades}} = \underbrace{C(A^T)}_{\dim r} \perp \underbrace{N(A)}_{\dim n-r} \qquad \underbrace{\mathbb{R}^m}_{\text{targets}} = \underbrace{C(A)}_{\dim r} \perp \underbrace{N(A^T)}_{\dim m-r}$$

Lesson 10 counted the dimensions (part 1 of the Fundamental Theorem). This lesson adds the angles (part 2): each pair splits its space into two perpendicular pieces.

Earlier idea	Seen as orthogonality
Do-nothing trade (Lesson 6)	perpendicular to every rule
Hidden link / reachable targets (Lesson 10)	reachable $\iff$ perpendicular to the hidden link
Arbitrage loop (Lesson 12)	Fredholm's certificate: $A^T y = 0, y^T b \neq 0$
Covariance $R^T R$ (Lesson 5)	a table of dot products; zero entries = uncorrelated

WHERE THIS GOES NEXT

When a target isn't reachable (or a risk can't be hedged exactly), drop a perpendicular from it onto the reachable plane. The foot of that perpendicular is the **best possible** target or hedge, and the leftover piece is orthogonal to everything you can trade. That's Lesson 14: projections.

7. Final memory table

Concept	Key fact	In markets
Orthogonal	$x^T y = 0$	market-neutral; uncorrelated returns
Length	$\|x\|^2 = x^T x$	risk of a demeaned return vector
Cosine	$x^T y / (\|x\|\|y\|)$	correlation
Row space ⊥ null space	$r + (n - r) = n$	do-nothing trades ⊥ every rule
Column space ⊥ left null space	$r + (m - r) = m$	reachable targets ⊥ hidden link
$A^T A$	invertible $\iff$ independent columns	overlap / covariance table
Fredholm	$Ax = b$, or $A^T y = 0$ with $y^T b = 1$	an arbitrage certificate

▼ Check yourself (MIT problem 16.1): multiply $x_1 - x_2 = 1$, $x_2 - x_3 = 1$, $x_1 - x_3 = 1$ by numbers so they add up to $0 = 1$.

$y = (1, 1, -1)$: $(x_1 - x_2) + (x_2 - x_3) - (x_1 - x_3) = 0$ on the left, $1 + 1 - 1 = 1$ on the right. So $0 = 1$: no solution. It's a spread loop: two one-step spreads of 1 each can't equal a direct spread of 1.